La Scuola d'Italia / Dr. Huson / IB Math 11
7 October 2026

Nickname: Solutions

1.10 Early Finishers: Sequences and Compound Interest

For students who finish the quiz early. A calculator is allowed. Show your working: write the formula you use, then substitute.

1. In an arithmetic sequence, $u_3 + u_8 = 48$ and $u_{12} = 5\,u_2$.

   (a) Find the first term u_1 and the common difference d.

$u_3 + u_8 = u_1 + d(3-1) + u_1 + d(8-1) = 48$

$2u_1 + 9d = 48 \longrightarrow 2\left(\frac{3}{2}d\right) + 9d = 48$

$12d = 48$

$d = 4$

$u_1 = 6$

$u_{12} = 5u_2$

$u_1 + d(12-1) = 5(u_1 + d(2-1))$

$u_1 + 11d = 5u_1 + 5d$

$6d = 4u_1 \quad u_1 = \frac{3}{2}d$

   (b) Find the number of terms of this sequence that are less than 500.

$u_n = 6 + 4(n-1) < 500$

$4n < 498$

$n < 124\frac{1}{2}$

124 terms

A different arithmetic sequence has first term 3 and common difference d, where $d \neq 0$. Its 1st, 2nd and 4th terms are the first three terms of a geometric sequence.

   (c) Find the value of d and the common ratio of the geometric sequence.

$u_1 = 3 \quad u_2 = 3 + d \quad u_4 = 3 + 3d$

$\frac{u_2}{u_1} = \frac{u_4}{u_2}$

$\frac{3+d}{3} = \frac{3+3d}{3+d}$

$(3+d)^2 = 3(3+3d)$

$d^2 + 6d + 9 = 9d + 9$

$d^2 - 3d = 0 \quad d \neq 0$

$d = 3$

$u_1 = 3 \quad r = 2$

$u_2 = 6$

$u_4 = 12$

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2. Sofia invests \$5000 in an account that pays a nominal annual interest rate of 4.8%, compounded monthly.

(a) Find the least number of whole years after which Sofia's investment is worth more than \$10 000.

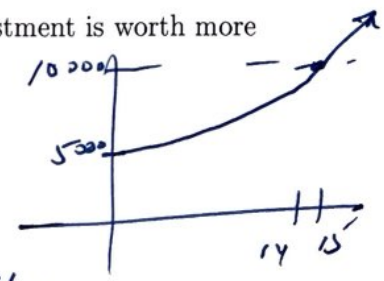
$FV = 5000 \cdot \left(1 + \frac{0.048}{12}\right)^{12t} > 10{,}000$

$(1.004)^{12t} > 2$

$12t > \log_{1.004} 2 = \not{t} \quad t \approx 173.6331\ldots$

$t > 14.469\ldots$

15 years



t	FV
14	9777.63...
15	10,257.47...

At the same time, Luca invests \$6000 in an account that pays a nominal annual interest rate of r%, compounded quarterly. After 10 years Luca's investment is worth \$9000.

(b) Find the value of r.

$FV = 6000\left(1 + \frac{r}{4}\right)^{4 \cdot 10} = 9000$

$\left(1 + \frac{r}{4}\right)^{40} = \frac{3}{2}$

$1 + \frac{r}{4} = \sqrt[40]{\frac{3}{2}} = 1.010881\ldots$

$r = 4.07527\ldots\ \%$

$\approx 4.08\ \%$

(c) Find the first whole year at the end of which Sofia's investment is worth more than Luca's.

$5000\left(1 + \frac{0.048}{12}\right)^{12t} > 6000\left(1 + \frac{0.048}{4}\right)^{4t}$

$\frac{5000\left(1 + \frac{0.048}{12}\right)^{12t}}{6000\left(1 + \frac{0.048}{4}\right)^{4t}} > \not{\$}\ \frac{6000}{5000} = 1.2$

$\left(\frac{\left(1 + \frac{0.048}{12}\right)^3}{1 + \frac{0.0408}{4}}\right)^{4t} > 1.2$

$1.0018294^{4t} > 1.2$

$t = \frac{1}{4}\log_{1.0018294} 1.2 = 24.938\ldots$

25 years



t	Luca	Sofia
24	15,877	15,784
25	16,534	16,560